

IPv4 and IPv6 Addresses

Overview: IPv4 and IPv6 are the two versions of the Internet Protocol currently in active use. IP addresses are generally allocated hierarchically: users receive addresses from ISPs, while ISPs obtain allocations through the appropriate Internet registry structure.

IANA: IANA allocates pools of unallocated IP address space to Regional Internet Registries (RIRs) according to applicable policies and documents protocol parameter assignments made by the IETF.

1. IPv4 Address

IPv4 was initially deployed on **1 January 1983** and remains widely deployed. An IPv4 address is a **32-bit** number normally represented as four 8-bit octets in dotted-decimal notation.

Example: 192.0.2.53

Key Characteristics

- **Structure:** 32 bits divided into four 8-bit octets.
- **Address Space:** 2^{32} , approximately 4.3 billion possible addresses.
- **Notation:** 0.0.0.0 through 255.255.255.255.
- **Network and Host Portions:** A subnet mask or prefix length identifies the network and host portions.

2. Historical IPv4 Address Classes

The following class ranges are part of the historical classful IPv4 model. Modern IPv4 networks primarily use CIDR and subnet prefixes rather than relying on Class A, B, and C.

Class	Range	Traditional Use
A	0.0.0.0 – 127.255.255.255	Large networks
B	128.0.0.0 – 191.255.255.255	Medium-sized networks
C	192.0.0.0 – 223.255.255.255	Smaller networks
D	224.0.0.0 – 239.255.255.255	Multicast
E	240.0.0.0 – 255.255.255.255	Reserved / experimental range

Loopback Address

The IPv4 range **127.0.0.0/8** is reserved for loopback communication. The commonly used loopback address is **127.0.0.1**, also known as *localhost*.

Subnet Mask

A subnet mask is a 32-bit value used with IPv4 to distinguish the network portion from the host portion of an address.

Network and Broadcast Addresses

- Examples of classful network-address patterns include n.0.0.0, n.n.0.0, and n.n.n.0.
- Examples of classful broadcast-address patterns include n.255.255.255, n.n.255.255, and n.n.n.255.

3. IPv4 Address Categories

Private IP Addresses

Private IPv4 addresses are used within local networks and are not globally routable on the public Internet. They are commonly used behind NAT.

- **10.0.0.0/8** — 10.0.0.0 to 10.255.255.255
- **172.16.0.0/12** — 172.16.0.0 to 172.31.255.255
- **192.168.0.0/16** — 192.168.0.0 to 192.168.255.255

Public IP Addresses

Public IPv4 addresses are globally routable addresses used for communication across the public Internet.

4. Finding an IPv4 Address

- **Windows:** Open Command Prompt and run `ipconfig`.
- **Linux:** Use `ip addr` or `ifconfig`.
- **Public IP:** A public-IP lookup service can display the public address visible from the Internet.

5. IPv6 Address

IPv6 deployment began in **1999**. IPv6 uses **128-bit** addresses represented in hexadecimal notation. Its much larger address space was designed to address the limitations of IPv4 address availability.

Core Structure

- **Bit Length:** 128 bits.
- **Notation:** Eight groups of four hexadecimal digits separated by colons.
- **Example:** 2001:0db8:85a3:0000:0000:8a2e:0370:7334
- **Address Space:** 2^{128} , approximately 3.4×10^{38} addresses.

6. IPv6 Notation Shortcuts

- **Leading Zeroes:** Leading zeroes within a hexadecimal group may be omitted.
- **Double Colon (::):** One consecutive sequence of all-zero groups can be compressed to :: once in an address.

Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334 → 2001:db8:85a3::8a2e:370:7334

7. Types of IPv6 Addresses

Unicast

A unicast address identifies a single interface. Packets sent to a unicast address are delivered to that interface. This represents one-to-one communication.

Multicast

A multicast address identifies a group of interfaces. Packets sent to the multicast address are delivered to interfaces that have joined the corresponding multicast group. This represents one-to-many communication.

IPv6 does not use broadcast addresses; multicast provides mechanisms for functions that would otherwise use broadcast.

Anycast

An anycast address is assigned to multiple interfaces. Packets sent to an anycast address are delivered to the interface selected as the nearest according to the routing system, commonly meaning the best routing metric.

8. IPv6 Unicast Addresses

Aggregatable Global Unicast

Global unicast addresses are globally routable IPv6 addresses designed to support hierarchical allocation and route aggregation. Current global unicast space is within **2000::/3**.

Link-Local

IPv6 link-local addresses use the prefix **FE80::/10**. They allow IPv6-enabled interfaces on the same local link to communicate without globally routable addresses. Routers do not forward link-local traffic beyond the local link.

Loopback

IPv4 uses **127.0.0.0/8** for loopback, with **127.0.0.1** commonly used as localhost. IPv6 uses **::1/128**.

Unspecified

The unspecified address contains all zero bits. IPv4 uses **0.0.0.0/32** and IPv6 uses **::/128**. It can be used when an interface does not yet have a valid address, including during processes such as DHCP.

Unique Local

Unique Local Addresses (ULA) are intended for private IPv6 communication within organizations or interconnected private networks. They use the **fc00::/7** range; locally assigned prefixes commonly begin with fd.

Embedded IPv4-in-IPv6

Some IPv6 transition mechanisms use IPv4 information embedded within an IPv6 representation. Such representations can support transition or tunneling mechanisms.

9. IPv6 Multicast

IPv6 multicast enables one-to-many communication. IPv6 multicast addresses use the prefix **ff00::/8**. A multicast packet has a unicast source address, while the destination can be a multicast group.

Well-known Multicast Addresses

Address	Purpose
FF02::1	All Nodes
FF02::2	All Routers
FF02::4	DVMRP Routers
FF02::5	All OSPFv3 Routers
FF02::6	OSPFv3 Designated Routers
FF02::A	All EIGRP (IPv6) Routers
FF02::D	All PIM Routers
FF02::12	VRRP
FF02::16	All MLDv2-capable Routers

10. Solicited-Node Multicast Address

A solicited-node multicast address is generated automatically for an IPv6 unicast address and is used for address resolution, Neighbor Discovery, and Duplicate Address Detection. The fixed prefix is **FF02::1:FF00:0/104**, combined with the last 24 bits of the corresponding IPv6 address.

11. Anycast Addresses

Anycast addresses are allocated from the unicast address space and therefore are not visually distinguishable from ordinary unicast addresses. The same address can be configured on multiple interfaces, allowing routing to deliver

packets to the selected nearest interface according to routing metrics.

Key Takeaway

IPv4: 32-bit, dotted-decimal, approximately 4.3 billion addresses.

IPv6: 128-bit, hexadecimal, vastly larger address space.

IPv4 private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.

IPv6: Supports unicast, multicast, and anycast; it does not use broadcast.

Memory aid: IPv4 = 32 bits + dotted decimal; IPv6 = 128 bits + hexadecimal + multicast-based local discovery.